Week 2 Practice Problem Solutions

3. File Extension Validator

Accept only pdf, docx, or zip extensions, regardless of case.

Java Solution

```
public class Main {
    static String validateFileExtension(String filename) {
        int dot = filename.lastIndexOf('.');
        if (dot < 0 || dot == filename.length() - 1) {
            return "Rejected - invalid file type";
        }
        String extension = filename.substring(dot + 1);
        if (extension.equalsIgnoreCase("pdf")
                || extension.equalsIgnoreCase("docx")
                || extension.equalsIgnoreCase("zip")) {
            return "Accepted";
        }
        return "Rejected - invalid file type";
    }

    public static void main(String[] args) {
        System.out.println(validateFileExtension("Assignment1.PDF"));
        System.out.println(validateFileExtension("notes.txt"));
    }
}
```

Sample Output

```
Accepted
Rejected - invalid file type
```

Notes

lastIndexOf() ensures the extension is taken from the final dot in the filename.